Endovascular thrombectomy for acute stroke: evolving eligibility criteria and adjunct therapies

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Endovascular thrombectomy is the standard treatment for ischaemic stroke, a leading cause of disability worldwide. Randomised trials in the past decade have expanded eligibility criteria to include broader patient populations, such as those with large ischaemic core stroke. However, many patients still have poor outcomes despite high rates of macrovascular reperfusion, underscoring the need for strategies that go beyond restoring large-vessel flow. Intra-arterial thrombolysis and cytoprotective agents are under investigation for their potential to reduce secondary complications during and after endovascular thrombectomy. These strategies target key mechanisms such as microvascular obstruction, excitotoxicity, oxidative stress, and inflammation. Their effectiveness depends on aligning the mechanism of action with the patient's underlying pathophysiology. Despite years of promising results limited by constrained clinical translation in cytoprotective trials, recent trials have provided hope and insights, helping to define crucial factors for translational success. Incorporating these lessons into refined inclusion criteria and future studies has the potential to transform our approach to caring for patients at the greatest risk of poor clinical outcomes, including severe disability or death.

Lancet Neurol 2026; 25: 61–76

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Introduction

In the past decade, endovascular thrombectomy has become the standard of care, transforming treatment for acute ischaemic stroke. Although initial trials of endovascular thrombectomy used strict inclusion criteria, randomised controlled trials (RCTs) throughout this decade have broadened eligibility, showing benefit in patient populations once considered poor candidates—including those with large core infarcts[1–6] and late time-window presentation.[7–9] Endovascular thrombectomy has also been proven safe and effective in low-income and middle-income countries, supporting its use in low-resource settings.[10] Despite the advances and high rates of macrovascular reperfusion, over half of patients treated with endovascular thrombectomy do not achieve functional independence, highlighting a substantial gap in the current treatment paradigm.[11] Strategies to narrow this gap include refining patient selection to identify those most likely to benefit and recognising those at higher risk of poor outcomes, improving the reperfusion quality at the microvascular level, and protecting vulnerable tissue from primary and secondary injury.

As ongoing trials continue to expand endovascular thrombectomy inclusion criteria and reshape clinical practice, attention is shifting toward identifying adjunctive strategies. These strategies include intra-arterial thrombolysis—the direct delivery of a clot-dissolving drug into the blocked artery via microcatheter, aiming to improve distal or microvascular reperfusion—and cytoprotective agents that target injury cascades triggered by both primary ischaemic and secondary reperfusion injury.[12] In this Review, we present recent developments, highlighting how biological targets align with the pathophysiological timeline of primary and secondary injury following ischaemic stroke. We also discuss why some trials have shown success and others might not have and propose important considerations for future trial design. This synthesis aims to inform neurologists, stroke specialists, and the broader neuroscience community about the evolving evidence for acute ischaemic stroke treatment and to inspire the design of next-generation trials and therapeutics.

Endovascular thrombectomy indications and patient eligibility

For years, endovascular thrombectomy was considered high risk and of little benefit in patients with large vessel occlusion, due to concerns about haemorrhagic transformation and poor functional recovery. Large vessel occlusion refers to a blockage in a major intracranial artery, such as the internal carotid artery (ICA) or proximal middle cerebral artery (MCA; figure 1), which typically results in large-territory infarction if untreated. This section reviews how evolving evidence is shaping endovascular thrombectomy use across a wider spectrum of patients with acute ischaemic stroke, including those historically considered poor candidates. It highlights where clear benefits have been shown and where uncertainties remain.

Patients with large ischaemic cores

Landmark randomised trials,[1–6] along with subsequent meta-analysis,[13] have shown significant functional benefit from endovascular thrombectomy in patients traditionally excluded due to large ischaemic cores. These RCTs focused on individuals with proximal large vessel occlusion in the anterior circulation, defined as Alberta Stroke Program Early CT Score (ASPECTS) 0–5 on non-contrast CT or diffusion-weighted MRI (DWI), or a core volume of ≥50 to ≥70 mL on CT perfusion (CTP; regional cerebral blood flow <30%) or MRI apparent diffusion coefficient less than 620×10^{-6} mm²/s. A study-level meta-analysis of six trials showed a mean improvement of 1·6 points on the 90-day modified Rankin Scale (mRS) with endovascular thrombectomy.[13] This improvement translated to a

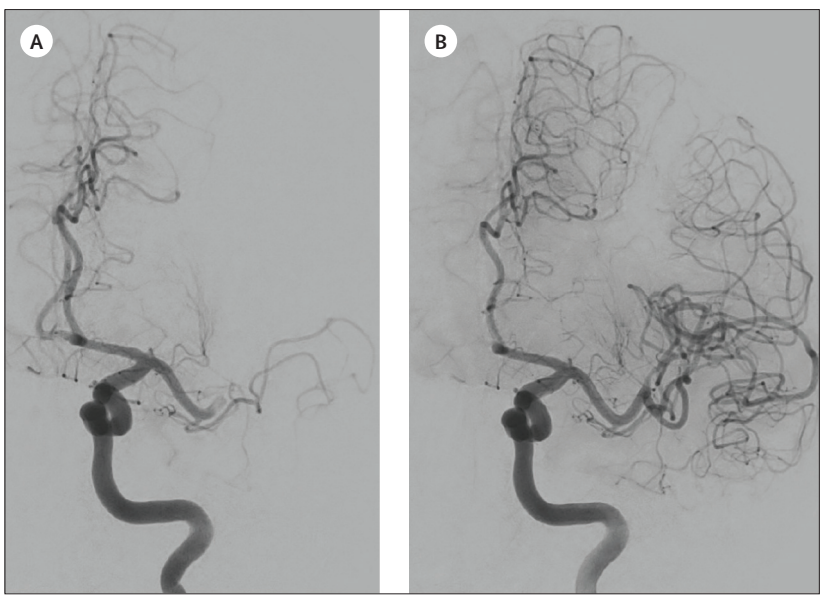


Figure 1: **Large vessel occlusion**
Example of a large vessel occlusion (distal M1) in a woman aged 87 years presenting with sudden right hemiparesis and aphasia. (A) Baseline digital subtraction angiogram of the left internal carotid artery shows abrupt cutoff of the distal M1 segment of the middle cerebral artery, with minimal opacification of downstream cortical branches. (B) Angiogram after endovascular thrombectomy (aspiration-only) demonstrates complete reperfusion of the left middle cerebral artery territory (expanded Thrombolysis in Cerebral Infarction score of 3).

doubling in the rate of functional independence (90-day mRS 0–2), increasing from 9% with medical management to 21% with endovascular thrombectomy and an increase in the proportion achieving mRS 0–3 from 22% to 37%, with higher risk of symptomatic intracerebral haemorrhage with endovascular thrombectomy (RR 1·7, 95% CI 1·1–2·7).[14] Observational data in patients with large ischaemic cores have also reported a numerical trend towards a higher risk of symptomatic intracerebral haemorrhage with endovascular thrombectomy, however, this finding was not statistically significant (odds ratio [OR] 1·83, 95% CI 0·95–3·55; p=0·07).[15]

Despite differences in imaging modalities used across studies, a consistent benefit (primarily defined by improved functional outcome) was observed in patients with ASPECTS 3–5 treated within the 0–6 h window, regardless of infarct size, time to treatment, or age.[16] This benefit appears to extend into the late time window, supported by findings from SELECT2,[3] ANGEL-ASPECT,[2] RESCUE-Japan LIMIT[1,17] (up to 24 h), and TENSION[4,18] (up to 12 h). However, a secondary analysis of ANGEL-ASPECT suggested that endovascular thrombectomy benefit might diminish with increasing time from symptom onset.[19] TESLA similarly reported a numerically weaker treatment effect (difference in mean utility-weighted mRS scores at 90 days) in patients treated later, although no statistically significant interaction with time was observed.[5] Trials further showed that patients with penumbral mismatch had better functional outcomes after endovascular thrombectomy than those without mismatch,[20,21] yet treatment benefits also occur for patients without mismatch,[3,22] indicating that simplified protocols using non-contrast CT and CT angiography might be sufficient in many large vessel occlusion scenarios.

Eligibility boundaries

As endovascular thrombectomy eligibility expands, several areas of uncertainty persist, one of which concerns patients with ASPECTS 0–2. Although a study-level meta-analysis of patients with ASPECTS 0–2 across six trials showed that endovascular thrombectomy within 0–6 h improved 90-day functional outcomes,[13,16] approximately 55% of these patients were from the LASTE trial, which allowed ASPECTS 0–2 (no upper infarct size limit), with over 80% assessed by DWI-MRI.[6] Excluding LASTE from the analysis eliminated statistical significance for endovascular thrombectomy benefit, likely due to reduced sample size and heterogeneity. Since DWI-MRI is more sensitive than non-contrast CT and tends to yield ASPECTS approximately one point lower,[23] ASPECTS 0–2 on MRI might reflect less severe ischaemia than on non-contrast CT—potentially explaining stronger evidence for endovascular thrombectomy benefit with MRI-based selection. There is no conclusive data on efficacy or safety for patients with ASPECTS 0–2 in the late time window (6–24 h), as LASTE included only patients eligible for treatment within 6·5 h of symptom onset. Furthermore, the benefit of endovascular thrombectomy in patients with large vessel occlusion generally appears to be time dependent,[5,19] likely reflecting evolving pathophysiological processes such as the development of vasogenic oedema. Thus, endovascular thrombectomy in patients with ASPECTS 0–2, particularly those selected by non-contrast CT and treated beyond early windows, should be approached with caution (figure 2A). Panel 1 further elaborates on imaging challenges and emerging markers, such CT hypodensity and net water uptake, which could help further refine patient selection and predict treatment response in this subgroup.

Another area of uncertainty involves endovascular thrombectomy for distal medium vessel occlusions in the distal M2, M3, anterior cerebral artery, and posterior cerebral artery territories, with recent randomised trials showing no significant difference between treatment and control groups.[30–32] DISCOUNT even showed inferiority, with higher rates of spontaneous intracerebral haemorrhage in the thrombectomy group compared with patients who did not undergo the procedure.[30] The reasons these trials failed to show a benefit of thrombectomy, despite expectations to the contrary, are likely multifactorial. Contributing factors include lower baseline National Institutes of Health Stroke Scale (NIHSS) scores (median 7–9), which can create a ceiling effect; high spontaneous recovery rates in the control group; anatomical challenges associated with tortuous distal vessels; heterogeneity in the thrombectomy devices used across trials, and so on.

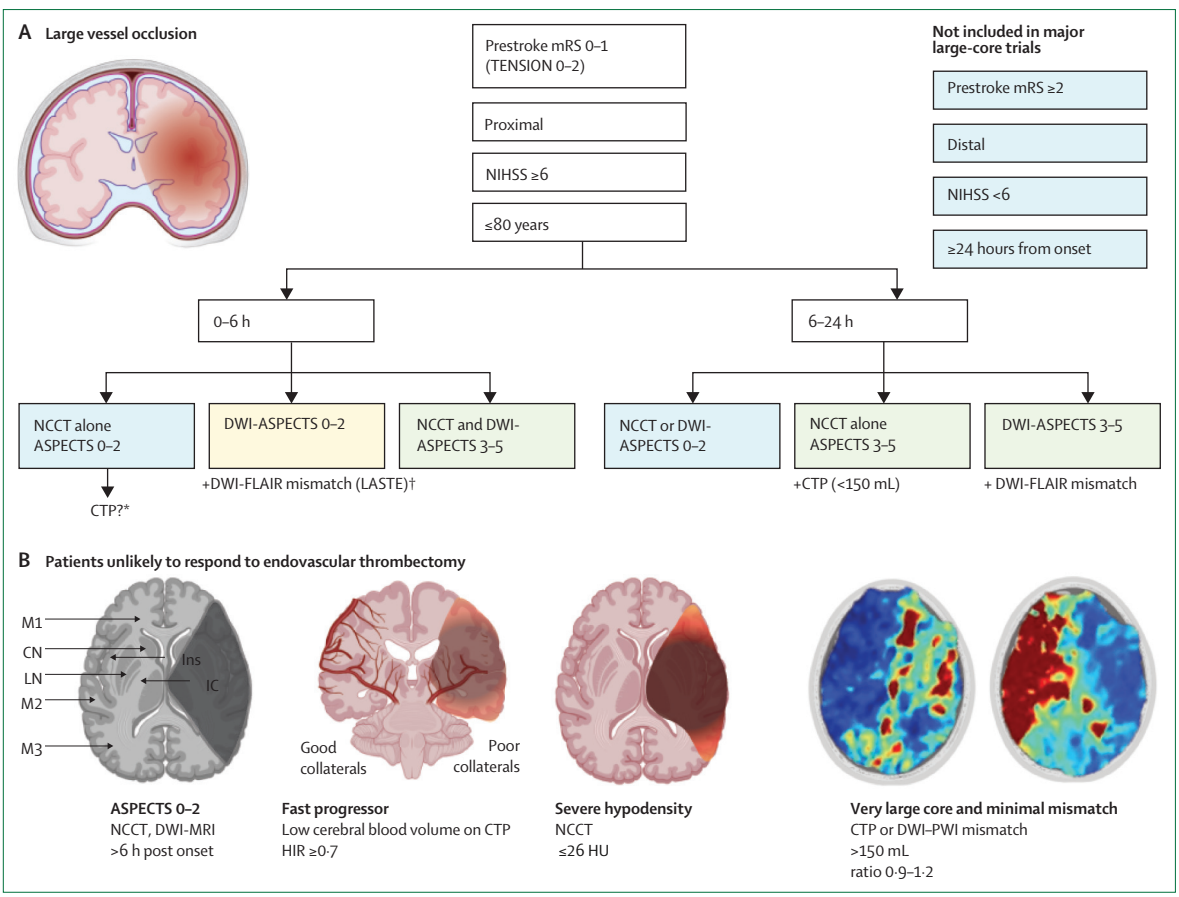


Figure 2: **Evidence framework for decision making in patients with large vessel occlusion undergoing endovascular thrombectomy**
This framework was produced by integrating information from RCTs and meta-analysis discussed in this Review. (A) Based on inclusion criteria from the six major trials investigating endovascular thrombectomy in patients with large ischaemic cores (eg, prestroke mRS 0–1 [or 0–2 in TENSION], proximal occlusion, NIHSS score ≥6, and age ≤80 years) and imaging-based confirmation of large vessel occlusion, patient groups shown in green are well supported by evidence for endovascular thrombectomy; yellow indicates those supported primarily by results from a single trial; and blue represents populations for whom evidence remains insufficient. Some of the groups not included in major large-core trials are involved in ongoing trials. Imaging modalities include NCCT or DWI-ASPECTS to estimate infarct size, and perfusion or mismatch imaging (CTP or DWI–FLAIR) in extended time windows or when the time of onset is unknown. *CTP might provide additional information in selected cases, but available data remain inconclusive in this subgroup. †DWI–FLAIR mismatch was used in LASTE[6] to identify patients within the 6 h window when symptom onset was unknown. (B) Potential non-responders to endovascular thrombectomy. ASPECTS=Alberta stroke program early CT score. CTP=CT perfusion. DWI-MRI=diffusion-weighted MRI. DWI–FLAIR mismatch=presence of hyperintensity on DWI-MRI without corresponding signal on fluid-attenuated inversion recovery, suggesting acute ischaemia. HIR=hypoperfusion intensity ratio. HU=Hounsfield units. mRS=modified Rankin Scale. NCCT=non-contrast CT. NIHSS=National Institutes of Health Stroke Scale. PWI=perfusion-weighted MRI. RCT=randomised controlled trial. Figure created with BioRender.com.

Compared with the HERMES data,[33] patients enrolled in the recent trials were older. Nonetheless, some subgroups, such as patients with proximal M2 occlusions, those with NIHSS scores higher than the median across trials, or those with intravenous thrombolysis failure, might still benefit from endovascular thrombectomy. In contrast, patients with more distal occlusions (eg, M3, anterior cerebral artery, and posterior cerebral artery) with low NIHSS scores, might have limited benefit, with procedural risks outweighing potential gains.

Still under-represented in endovascular thrombectomy trials are patients with mild deficits (NIHSS score ≤5) and proximal large vessel occlusions, a group currently under investigation in the ENDOLOW (NCT04167527) and MOSTE (NCT03796468) trials. Another population for which endovascular thrombectomy evidence remains limited involves patients with pre-existing disability (mRS ≥3). All major RCTs included in this Review excluded these individuals. Although observational studies suggest that some patients might return to their prestroke baseline, outcomes remain heterogeneous.[34] Additional studies are currently underway (eg, through the Patient-centred Outcomes Research Institute project). Although age alone should not be used as an exclusion criterion,[35] there are also few data on endovascular thrombectomy safety and efficacy in older patients (>85 years).

Despite recent encouraging data showing the benefit of endovascular thrombectomy in patients with large core infarcts, a substantial subset of patients with large vessel occlusion remain at high risk for poor outcomes. Besides the established subgroups of patients unlikely to respond

For more on the **Patient-centred Outcomes Research Institute project** see https://www.pcori.org

Panel 1: **Challenges and new imaging markers in patients with large vessel occlusion**

ASPECTS variability

Trials have emphasised variability in Alberta Stroke Program Early CT Score (ASPECTS) interpretation, especially across different imaging modalities.[1–6] Scoring accuracy is highly dependent on CT quality and window-level settings. Loss of gray–white matter differentiation, particularly in cortical regions (M1–M6), is a critical ischaemia marker but is susceptible to oedema and technical factors, whereas subcortical regions (caudate, lentiform nucleus, and internal capsule) typically show earlier and more homogeneous hypodensity. Diffusion-weighted MRI-ASPECTS might overestimate the core.[24] A large core, as defined solely by ASPECTS less than or equal to 5, might not be as reliable as once assumed, as this definition masks the underlying tissue heterogeneity. Similarly, the upper benefit threshold based on infarct volume remains debated.[13,25]

CT hypodensity and net water uptake

Given the limitations of ASPECTS, the hypodensity on non-contrast CT and net water uptake emerge as important markers. Yogendrakumar et al[26] identified a threshold of 26 mL of severe hypodensity (characterised by marked hypoattenuation and sulcal effacement) as predictive of "no significant endovascular thrombectomy benefit". Volumes exceeding this threshold were strongly associated with higher risks of malignant oedema, hemicraniectomy, and poor functional outcomes. Similarly, higher net water uptake on admission CT correlates with malignant oedema, haemorrhagic transformation, and poor outcomes, particularly in patients with low ASPECTS.[27,28] Net water uptake levels above approximately 12% indicate advanced injury and early oedema, identifying those unlikely to benefit from endovascular thrombectomy despite recanalisation.[29]

to endovascular thrombectomy—those with poor baseline functional status (modified Rankin Scale scores >2), very late presentations (>24 h from onset), severe brain swelling or injury (eg, midline shift, herniation, or mass effect), bilateral strokes, and tandem or multiple intracranial occlusions—recent evidence suggests several imaging-defined non-responder profiles (figure 2B). Unsuccessful reperfusion (modified Thrombolysis in Cerebral Infarction [mTICI] 0–2a) remains arguably the strongest predictor of poor outcome, and patients with low ASPECTS who do not have mTICI scores of at least 2b have the highest rates of poor outcomes.[36,37]

Stroke systems of care should prepare for a roughly 15–20% increase in endovascular thrombectomy volume, as approximately one in five large vessel occlusion strokes will now meet eligibility criteria for the procedure.[38] This anticipated increase further emphasises the need to extend the benefits of endovascular thrombectomy through cytoprotection and other adjunctive therapies, especially for patients who might stand to gain the most from such approaches.

Adjunct therapies

Many patients treated with endovascular thrombectomy do not reach meaningful functional recovery—either because blood flow is only partly restored or, in some cases, poor outcomes occur despite complete reperfusion. These poor outcomes are due to persistent microvascular obstruction or tissue injury caused by the initial deprivation of oxygen and nutrients (ischaemia) and the sudden restoration of blood flow (reperfusion injury). This barrier to recovery has driven the search for adjunctive therapies aimed at enhancing reperfusion and mitigating ischaemia–reperfusion injury. The following section reviews intra-arterial thrombolysis and cytoprotective agents as potential strategies to extend the benefits of endovascular thrombectomy.

Intra-arterial thrombolysis

Despite high rates of successful reperfusion following endovascular thrombectomy, approximately 50% of patients do not reach functional independence at 90 days.[39,40] This discrepancy highlights the growing recognition that angiographic success (defined as an expanded Thrombolysis in Celebral Infarction [eTICI] score of 2b–3) does not always translate to effective tissue-level reperfusion, likely due to persistent microvascular obstruction known as no-reflow.[41,42] Intra-arterial thrombolysis has re-emerged as a promising adjunctive strategy to address this limitation by targeting residual thrombi in the distal microcirculation (figure 3A).[43]

The CHOICE phase 2 trial,[44] conducted in Spain, evaluated intra-arterial alteplase following successful endovascular thrombectomy and reported an approximately 18% absolute increase in the proportion of patients achieving mRS of 0–1 at 90 days. However, CHOICE was a single-region study that was terminated early due to a placebo shortage. Subsequent trials explored the approach in patients with near-complete to complete reperfusion (eTICI 2c–3), with intra-arterial urokinase (100 000 IU) in POST-UK,[45] and intra-arterial tenecteplase at 0·0625 mg/kg (one-fourth the standard intravenous dose) in POST-TNK.[46] Although post-procedural intra-arterial thrombolysis appeared to show slightly better outcomes (~5% across trials), the difference was not statistically significant, and neither POST-UK nor POST-TNK met their primary endpoints of excellent functional outcome at 90 days, defined as an mRS score of 0 or 1. The ATTENTION-IA trial[47] tested intra-arterial tenecteplase after successful endovascular thrombectomy in patients with posterior circulation stroke but did not show a significant increase in excellent recovery (34·6% *vs* 26%; adjusted risk ratio [RR] 1·36, 95% CI 0·92–2·01).

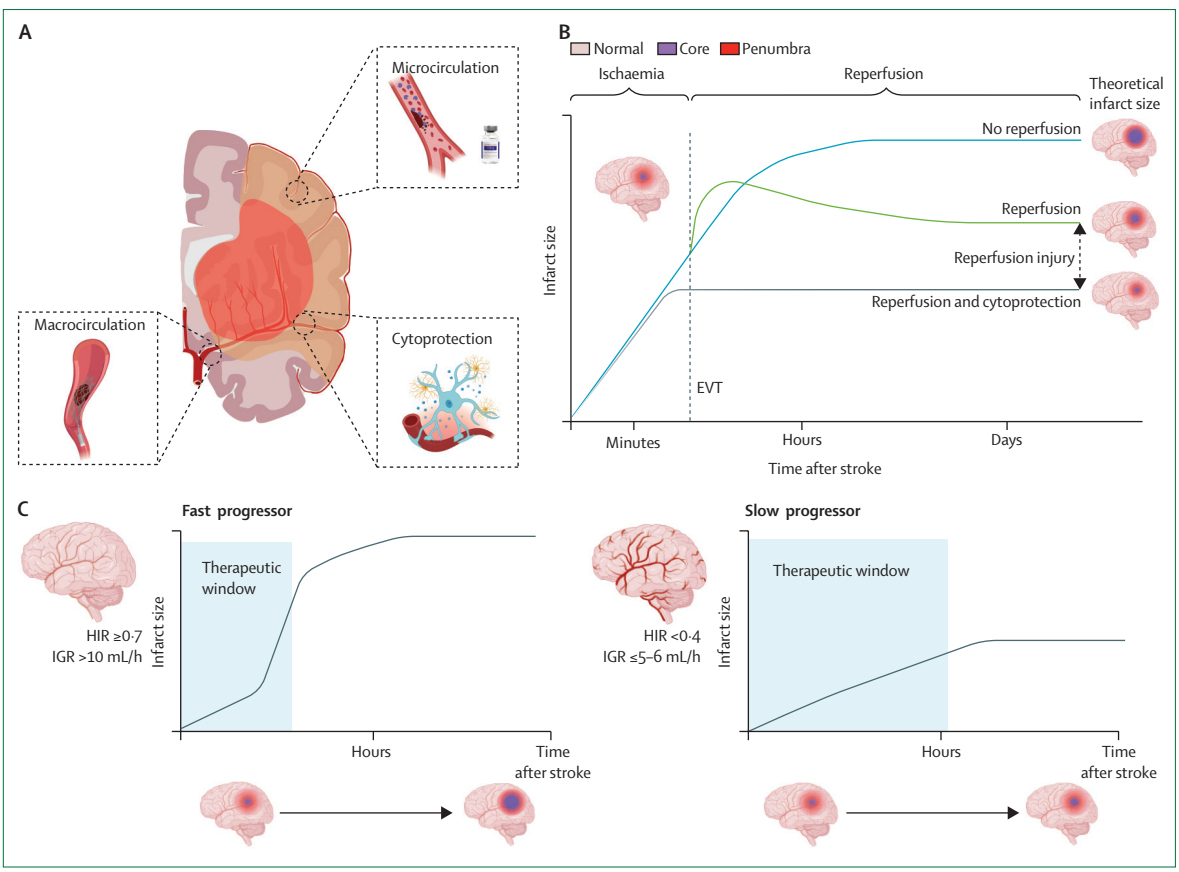


Figure 3: **Strategies and factors influencing outcomes and therapeutic response in arterial ischaemic stroke**
(A) Strategies to improve outcomes in patients with large vessel occlusion include targeting macrocirculation (through EVT) and microcirculation (through IVT), along with cytoprotective approaches aimed at mitigating both primary and secondary injury. (B) Infarct size evolution over time. In the absence of reperfusion (blue line), infarct size continues to expand toward the theoretical maximum. Reperfusion (green line) halts core progression but might introduce secondary reperfusion injury, limiting recovery. The addition of cytoprotective therapies (grey line) aims to mitigate ischaemia–reperfusion injury, preserving more viable tissue. EVT is indicated at the transition between ischaemia and reperfusion phases. (C) Infarct growth phenotypes: fast progressors (IGR >10 mL/h, HIR ≥0·7) rapidly lose penumbra, leaving a narrow therapeutic window; slow progressors (IGR ≤5–6 mL/h, HIR <0·4) maintain penumbra longer, allowing for an extended window for reperfusion and cytoprotective therapies. EVT=endovascular thrombectomy. HIR=hypoperfusion intensity ratio. IGR=infarct growth rate. IVT=intravenous thrombolysis. Figure created with BioRender.com.

The DATE trial,[48] a phase 1/2a dose-escalation study, assessed intra-arterial tenecteplase at three doses: one-eighth, one-fourth, and one-half of the standard intravenous dose. The doses of 0·0313 mg/kg and 0·0625 mg/kg showed an adequate safety profile, whereas the higher dose of 0·125 mg/kg exhibited safety concerns due to a high rate of symptomatic intracerebral haemorrhage. However, the two lower, safer doses did not show significant improvement in functional outcomes at 90 days, as the sample size was small and the trial was not designed to show efficacy. In contrast, the ANGEL-TNK trial[49] evaluated intra-arterial tenecteplase at 0·125 mg/kg and showed a statistically significant difference in excellent functional outcomes (mRS 0–1) at 90 days, with 40·5% of patients treated with intra-arterial tenecteplase reaching this score compared with 26·4% of patients who did not receive the treatment. At the 2025 International Stroke Conference, the PEARL trial[50] reported their preliminary findings that intra-arterial alteplase (0·225 mg/kg, one-fourth the intravenous dose) improved 90-day excellent outcomes (mRS 0–1) without increasing symptomatic intracerebral haemorrhage or mortality. A meta-analysis of seven trials showed that endovascular thrombectomy with intra-arterial thrombolysis significantly improved 90-day functional outcomes without increasing the risk of symptomatic intracerebral haemorrhage or mortality.[51] However, the increase in radiological intracerebral haemorrhage at 48 h reported in one of the included trials, POST-TNK, warrants caution. The trials varied in design, thrombolytic agents, and often enrolled small subgroups of patients receiving intravenous thrombolysis. Additionally, the predominance of single-country trials raises concerns about the generalisability of the findings.

ANGEL-TNK[49] and PEARL[50] enrolled patients with anterior large vessel occlusion and eTICI 2b–3 reperfusion scores, broadening the criteria of POST-UK[45] and

POST-TNK[46]. This group showed a stronger signal for benefit from intra-arterial thrombolysis compared with those with eTICI 2c–3, reinforcing the possibility that intra-arterial thrombolysis might not only aid in addressing microvascular obstruction, but also augment macrovascular reperfusion.[52] Patients with eTICI 2a or less were not included, and as such, the use of intra-arterial thrombolysis as a rescue strategy in cases of unsuccessful endovascular thrombectomy remains understudied. The ongoing TECNO trial (NCT05499832) and EXTEND AGNES (NCT05892510) are evaluating intra-arterial tenecteplase for unsuccessful, incomplete, or near-complete reperfusion (TICI 0–2c) in anterior and posterior circulation strokes. Other trials, including IA-SUCCESS (NCT06768138), IAT-TOP (NCT05897554), and INSIST IT[53] (NCT05657457) will further inform our understanding of the efficacy and safety of adjunctive thrombolysis with endovascular thrombectomy.

See Online for appendix

Given the efficacy signals across various dose regimens and thrombolytic agents reported (appendix p 6), current findings support further exploration of intra-arterial thrombolysis as an adjunct to endovascular thrombectomy. However, the results also indicate the importance of determining the optimal thrombolytic dose and timing and refining patient selection criteria—including whether the number of endovascular thrombectomy passes, infarct size, stroke location, or previous use of intravenous thrombolysis should influence decision making. Ongoing studies should provide further insight on whether intra-arterial thrombolysis could be applied as a rescue strategy in cases of unsuccessful recanalisation. Other strategies to improve tissue-level reperfusion, including novel thrombolytics, reperfusion enhancement approaches, and periprocedural antithrombotics, are presented in the appendix (p 5).

Cytoprotective strategies

Many adjunctive interventions have been designed to target one or more of the pathways outlined in panel 2. The following section highlights recent cytoprotective trials (table).

Excitotoxicity

Nerinetide is a peptide developed to counteract NMDA receptor-mediated excitotoxicity by disrupting the interaction between postsynaptic density protein 95 (PSD-95) and glutamate ionotropic receptor NMDA type subunit 2B (GluN2B), limiting calcium influx and nitric oxide-induced neuronal injury.[70] Its clinical potential was first explored in ESCAPE-NA1,[71] a phase 3 trial enrolling patients with acute ischaemic stroke within 12 h of symptom onset who were eligible for endovascular thrombectomy. Although the trial showed no overall 90-day functional benefit, subgroup analysis suggested improved outcomes without previous intravenous alteplase, raising the possibility that alteplase degrades nerinetide.[71] The ESCAPE-NEXT[69] trial excluded patients treated with intravenous thrombolysis and re-evaluated nerinetide in the context of endovascular thrombectomy alone, but again showed no clinical benefit, highlighting the narrow therapeutic window. In many cases, nerinetide was administered shortly before rapid and complete reperfusion, often within 30–45 min of reaching eTICI 2c–3, leaving little time for the drug to act.[69] The enrolled population also included many slow progressors with good collateral circulation, in whom rapid reperfusion might make it harder to detect incremental cytoprotective effects.

The phase 2 FRONTIER[72] trial administered nerinetide in the prehospital setting within 3 h of symptom onset. Although the primary outcome was non-significant, subgroup analyses pointed to a possible benefit among patients with confirmed ischaemic stroke who later received reperfusion.[72] A meta-analysis of patients treated within 3 h who had successful reperfusion showed that nerinetide significantly improved 90-day functional independence compared with placebo.[73] A next-generation PSD-95 inhibitor (NoNO-42), not susceptible to thrombolytic degradation, is under evaluation in ACT-42 (NCT06403267). This phase 2b trial is enrolling up to 600 adults aged 45–90 years within 3 h of stroke onset, all of whom are eligible for intravenous thrombolysis, with or without endovascular thrombectomy. Participants are randomly assigned (1:1) to NoNO-42 or placebo, with primary efficacy outcomes assessed at 90 days and safety outcomes at 30 days.

Blood–brain barrier disruption and cytotoxic oedema

Glyburide (also known as glibenclamide) targets the sulfonylurea receptor 1–transient receptor potential melastatin 4 (SUR1–TRPM4) channel, a key mediator of cytotoxic oedema following large hemispheric infarction.[74] Two clinical trials evaluated its efficacy in this setting. The phase 2 GAMES-RP trial[75] enrolled 86 patients aged 18–80 years with DWI-MRI confirmed lesion volumes of 82–300 mL within 10 h of symptom onset. Although the primary composite endpoint (90-day mRS 0–4 without decompressive craniectomy) was not met, glyburide significantly reduced midline shift, plasma MMP-9 levels, and cerebral oedema.[75] Patients 70 years or younger also showed significantly lower mortality at 90 days and 12 months, suggesting possible long-term benefit in this subgroup.[76]

Building on these findings, the phase 3 CHARM trial,[77] conducted across 21 countries, enrolled 535 patients with large hemispheric infarction (ASPECTS 1–5 or ischaemic core 80–300 mL, confirmed by CT or MRI) within 10 h of symptom onset. Terminated early due to COVID-related slow enrolment, CHARM's primary outcome—an ordinal shift in 90-day mRS—showed no significant group difference.[77] This outcome likely reflected broader and more heterogeneous imaging criteria (including three different imaging modalities) than in GAMES-RP, missing midline shift data in severe cases, and variability

Panel 2: **Targets and injury mechanisms in acute ischaemic stroke**

Restoring blood flow is essential to salvage at-risk tissue but also initiates a cascade of secondary injury. In the modern era of stroke management, where reperfusion therapies are routinely applied, there are two principal injuries: the initial ischaemic injury from oxygen and glucose deprivation, and the reperfusion-induced secondary injury.[54] These mechanisms are dynamic, overlapping (figure 4), and their timeline can vary between fast and slow progressors and between successful and unsuccessful reperfusion.

Primary ischaemic injury

Within minutes of occlusion, neurons, endothelial cells, and glial cells undergo metabolic collapse due to ATP depletion and Na^+/K^+-ATPase pump dysfunction. Ionic imbalance results in cytotoxic oedema with sodium, chloride, and calcium accumulation (figure 4A).[55] Hypoxia further impairs cyclic adenosine monophosphate signalling, weakening endothelial tight junctions and increasing blood–brain barrier permeability.[56] Excess glutamate overstimulates NMDA receptors, driving calcium influx, mitochondrial dysfunction, and excitotoxic neuronal death.[56] Reduced nitric oxide synthesis limits vasodilation and promotes leukocyte adhesion.[57]

Within the first hours of occulsion, astrocytes swell, impairing glutamate uptake and lactate delivery needed to support neuronal metabolism.[56] Their glutamate transporters might reverse, increasing extracellular glutamate levels and worsening excitotoxicity.[58] Microglia within the core degenerate, whereas those in the surrounding penumbra remain metabolically viable but are primed for activation.[59]

Reperfusion injury

Reintroduction of oxygen into a metabolically compromised environment can exacerbate injury—the so-called oxygen paradox.[60] Reactivated mitochondria generate large quantities of reactive oxygen species (ROS), particularly via reverse electron transport at complex I, driven by accumulated succinate.[56] This process leads to lipid peroxidation, oxidative damage to mitochondrial proteins, and opening of the mitochondrial permeability transition pore (mPTP; figure 4C).[61] Simultaneously, reactive nitrogen species (RNS) such as peroxynitrite—formed from superoxide (O_2^-) and nitric oxide—damage cellular proteins and DNA, promoting both apoptotic and necrotic cell death.[62]

Inflammatory cascade

Between 1 h and 6 h post-reperfusion, inflammation becomes central. Damage-associated molecular patterns (DAMPs) activate Toll-like receptors (TLR2 and TLR4) on microglia and endothelium.[63] This activation induces production of pro-inflammatory cytokines, matrix metalloproteinases (MMPs), and upregulation of NADPH oxidase 2 (NOX2) and inducible nitric oxide synthase (NOS2 or iNOS), elevating ROS levels and exacerbating blood–brain barrier disruption.[64] The endothelium also secretes pro-inflammatory molecules and expresses adhesion molecules such as P-selectin, facilitating leukocyte adhesion and infiltration.[65] Neutrophils, among the earliest responders, express NOS2 and, together with activated microglia, intensify oxidative damage, worsening oedema and facilitating its progression to vasogenic oedema or even haemorrhagic transformation (figure 4A).[66]

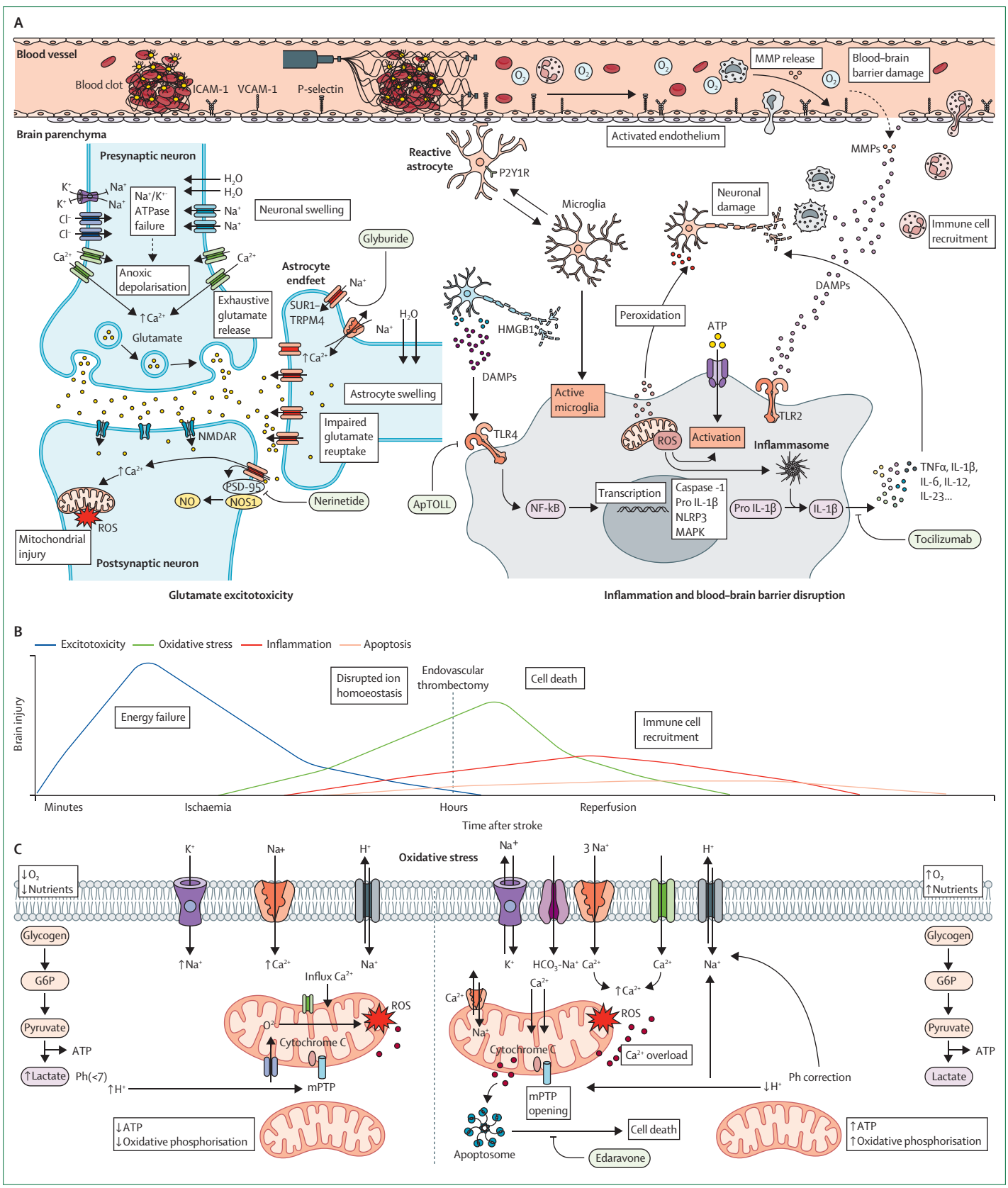
Delayed injury and repair

Between 6 h and 72 h, ROS and RNS levels exhibit biphasic peaks due to ongoing mitochondrial dysfunction and inflammation.[67] Continued mPTP opening sustains apoptotic signalling and p53–cyclophilin D-mediated necrosis.[68] Despite this continued injury, reparative mechanisms gradually prevail. Microglia adopt anti-inflammatory phenotype, producing IL-10 and IL-4 and transforming growth factor beta, aiding in debris clearance and tissue repair. Neurotrophic factors including GDNF and BDNF support neuronal survival and restore homoeostasis.[69]

in standard of care across a wide international network. Early termination left the trial underpowered, and changes in practice since GAMES-RP, including increased endovascular thrombectomy use, might have further influenced the biological context for oedema and treatment effect. However, in a subgroup with ischaemic cores less than 125 mL, particularly among patients treated with endovascular thrombectomy, intravenous glyburide was associated with improved mRS outcomes, reduced decompressive craniectomies, midline shift, and mortality.[78] Favourable outcomes in endovascular thrombectomy-treated patients likely reflects a more optimal window for modulating injury processes, particularly oedema formation. In moderately large, but not overwhelming infarcts, the pathophysiological cascade leading to malignant oedema might still be pharmacologically tractable. The apparent volume threshold of approximately 125 mL identified in the CHARM trial likely delineates a tipping point beyond which the extent of irreversible injury and cytoskeletal disruption becomes too extensive for anti-oedema therapies to exert meaningful benefit.[78] Further trials are needed to evaluate glyburide in the optimal target population, stratified by baseline ischaemic core thresholds and endovascular thrombectomy status.

Oxidative stress and inflammation

Edaravone dexborneol is a combination of edaravone, a free radical scavenger, and dexborneol, a monoterpenoid anti-inflammatory compound. Together, they target oxidative stress, excitotoxicity, calcium overload, and inflammation.[79,80] Preclinical studies have shown that edaravone dexborneol downregulates ferroptosis, offering potential protection against reperfusion injury.[79] The phase 3 TASTE-SL[81] trial tested sublingual edaravone dexborneol in 914 patients with acute ischaemic stroke


A
Blood vessel
Blood clot
ICAM-1
VCAM-1
P-selectin
MMP release
Blood–brain barrier damage
Activated endothelium
Brain parenchyma
Presynaptic neuron
Na+/K+ ATPase failure
Neuronal swelling
Anoxic depolarisation
Exhaustive glutamate release
Glutamate
NMDAR
PSD-95
NOS1
NO
ROS
Mitochondrial injury
Postsynaptic neuron
Glyburide
Astrocyte endfeet
SUR1-TRPM4
Astrocyte swelling
Impaired glutamate reuptake
Nerinetide
Glutamate excitotoxicity
Reactive astrocyte
P2Y1R
Microglia
HMGB1
DAMPs
TLR4
ApTOLL
Active microglia
NF-kB
Transcription
Caspase -1
Pro IL-1β
NLRP3
MAPK
Pro IL-1β
IL-1β
Inflammasome
ROS
Activation
ATP
TLR2
Peroxidation
Neuronal damage
MMPs
Immune cell recruitment
TNFα, IL-1β, IL-6, IL-12, IL-23...
Tocilizumab
Inflammation and blood–brain barrier disruption
B
Excitotoxicity
Oxidative stress
Inflammation
Apoptosis
Brain injury
Energy failure
Disrupted ion homoeostasis
Endovascular thrombectomy
Cell death
Immune cell recruitment
Minutes
Ischaemia
Hours
Reperfusion
Time after stroke
C
Oxidative stress
↓O2 ↓Nutrients
↑O2 ↑Nutrients
Glycogen
G6P
Pyruvate
ATP
Lactate
Ph(<7)
↑H+
mPTP
Influx Ca2+
Cytochrome C
ROS
↓ATP ↓Oxidative phosphorisation
Ca2+ overload
mPTP opening
Apoptosome
Cell death
Edaravone
Ph correction
↓H+
↑ATP ↑Oxidative phosphorisation

treated within 48 h of symptom onset. Patients received edaravone dexborneol twice daily for 14 days. TASTE-SL showed a significant increase in excellent outcomes at 90 days (mRS 0–1), without an increase in serious adverse events.[81] The rapid disintegration and absorption through the sublingual route offer a practical strategy for early or even prehospital administration.

To assess edaravone dexborneol in patients undergoing endovascular thrombectomy, a phase 2 randomised trial enrolled 200 patients with anterior-circulation large vessel occlusion who had successful reperfusion.[82] Patients received intravenous edaravone dexborneol or placebo twice daily for 12 days. Although the primary outcome (mRS 0–2 at 90 days) was not statistically significant, the edarabone dexborneol group numerically appeared to have reduced infarct volume and improved early neurological recovery, but these findings were not statistically significant either. A time-of-day interaction suggested greater efficacy when edaravone dexborneol was administered during the daytime, indicating potential circadian influences on drug response.[82] The ongoing phase 3 TASTE-2 trial is enrolling patients with anterior-circulation large vessel occlusion (NIHSS 6–25, ASPECTS 6–10) within 24 h of symptom onset, with treatment continued for 10–14 days. The primary outcome is mRS 0–2 at 90 days, evaluating whether edaravone dexborneol enhances outcomes when added to standard reperfusion therapies.[83]

The ERASE-STROKE (NCT06648304) trial, presented at the European Stroke Organisation Conference (ESOC) 2025, was a phase 3 study evaluating oral edaravone dexborneol (TTYP01) in patients with acute ischaemic stroke across 40 Chinese stroke centres. Randomly assigned within 24 h of stroke onset, patients received TTYP01 or placebo for 28 days. The primary endpoint, mRS 0–1 at 90 days, was significantly improved in the TTYP01 group. However, mRS 0–2 did not reach statistical significance. Among secondary endpoints, only an NIHSS score 4 or more at day 28 was significantly improved. TTYP01 was not linked to higher intracerebral haemorrhage or mortality. The study excluded patients undergoing reperfusion or with cardioembolic stroke. A follow-up trial, ERASE-STROKE 2, is planned to include broader populations and assess use alongside reperfusion therapies.

Tocilizumab, a monoclonal antibody against the IL-6 receptor, has emerged as a potential cytoprotective agent in acute ischaemic stroke by targeting inflammation-mediated secondary injury.[84] The IRIS trial,[85] also presented at 2025 ESOC, was a phase 2 trial evaluating the safety and efficacy of tocilizumab in patients with acute ischaemic stroke due to anterior-circulation large vessel occlusion undergoing endovascular thrombectomy. Conducted across six stroke centres in China, the trial enrolled 108 patients who were randomly assigned (1:1) to receive either tocilizumab or placebo within 1 h after randomisation. The primary outcome showed that tocilizumab significantly reduced ischaemic core growth at 72 h compared with placebo. Safety outcomes were favourable when compared with placebo, with a significantly lower rate of symptomatic intracerebral haemorrhage and no significant differences in mortality or serious adverse events. Secondary outcomes, including early neurological improvement, final reperfusion grade, and functional independence at 90 days (mRS 0–2), did not differ significantly between the groups. Overall, findings suggest a cytoprotective effect of tocilizumab, supporting the need for larger confirmatory trials.

Figure 4: **Temporal evolution and therapeutic targets in ischaemia–reperfusion with cytoprotective agents currently in clinical trials**
(A) Following occlusion, abrupt cessation of O_2 and glucose leads to ATP depletion, disabling the Na^+/K^+-ATPase pump and causing membrane depolarisation and neuronal swelling. Excess presynaptic glutamate overstimulates NMDA receptors, particularly extrasynaptic *GRIN2B* subtypes, leading to Ca^{2+} influx, ROS generation, and mitochondrial failure; nerinetide targets this excitotoxic injury via PSD-95–GluN2B interaction. Astrocytic SUR1–TRPM4 channels mediate Na^+-coupled H_2O influx, promoting cytotoxic oedema; intravenous glyburide targets this pathway. Thrombus removal by endovascular thrombectomy restores flow but activates endothelium, increasing the expression of cell-adhesion molecules (ICAM-1, VCAM-1, and P-selectin). This sudden reintroduction of O_2 further drives oxidative stress, blood–brain barrier disruption, and leukocyte recruitment. DAMPs (eg, ATP and HMGB1) released by damaged neurons activate TLR2 and TLR4 on microglia and endothelium, amplifying NF-κB signalling. ApTOLL (TLR4 antagonist) and tocilizumab (IL-6 receptor blocker) act at these checkpoints. Activated microglia release cytokines and chemokines to recruit leukocytes, whose secretion of MMPs further compromises endothelial integrity. (B) Timeline of dominant injury mechanisms following occlusion and reperfusion. Energy failure and excitotoxicity peak early; oxidative stress rises rapidly after blood-flow restoration and gradually falls. Inflammation and apoptotic signalling accumulate over hours and then recede as reparative processes emerge. The dashed line marks the timing of endovascular thrombectomy. (C) Ionic shift and mitochondrial derangements underlying oxidative injury. Ischaemia reduces O_2 and nutrient availability, forcing anaerobic glycolysis, lactate accumulation, and intracellular acidosis. Reperfusion induces Ca^{2+} overload and sustained mPTP opening, elevating ROS, RNS, and lipid peroxidation, leading to cell death. Investigational agent edaravone, a free-radical scavenger, attenuates mitochondrial oxidative damage and supports membrane integrity. Coloured curves and arrows illustrate sequence and overlap of injury mechanisms, not quantitative proportions. AIS=acute ischaemic stroke. DAMPs=damage-associated molecular patterns. G6P=glucose-6-phosphate. GluN2B=glutamate ionotropic receptor NMDA type subunit 2B. *GRIN2B*=glutamate ionotropic receptor NMDA type subunit 2B; HMGB1=high-mobility group box 1. ICAM-1=intercellular adhesion molecule 1. MMPs=matrix metalloproteinases. mPTP=mitochondrial permeability transition pore. NF-κB=nuclear factor kappa-light-chain-enhancer of activated B cells. PSD-95=postsynaptic density protein 95. RNS=reactive nitrogen species. ROS=reactive oxygen species. SUR1–TRPM4=sulfonylurea receptor 1–transient receptor potential melastatin 4 channel. TLR=toll-like receptor. VCAM-1=vascular cell adhesion molecule 1. Figure created with BioRender.com.

Multimechanisms of ischaemic reperfusion injury

DL-3-n-butylphthalide (NBP) is a synthetic compound derived from celery seed extract, with cytoprotective effects via multiple mechanisms, including improvement of microcirculation, inhibition of mitochondrial dysfunction, reduction of oxidative stress, and attenuation of neuronal apoptosis.[86] Although NBP has been widely used in China for ischaemic stroke, few large-scale confirmatory trials had been done. BAST, a phase 3 trial, evaluated the efficacy and safety of NBP in patients with acute ischaemic stroke following reperfusion.[87] The study enrolled 1216 adult patients who began the NBP within 6 h of stroke onset

	Trial name	Phase	Eligibility criteria	Time window	Administration timing	Primary outcome	Status	Results
Excitotoxicity								
Nerinetide: PSD-95–NMDAR disruption	ESCAPE-NA1	3	LVO undergoing EVT; NCCT-ASPECTS >4; moderate-to-good collaterals on mCTA; NIHSS >5	Within 12 h	Before EVT (some with alteplase)	mRS score of 0–2 at day 90	Completed	Overall non-significant, significant benefit in no-thrombolysis subgroup
Nerinetide	ESCAPE-NEXT	3	LVO undergoing EVT; NCCT-ASPECTS >4; NIHSS >5	Within 12 h	Before the end of EVT (no thrombolysis)	mRS score of 0–2 at day 90	Completed	Overall non-significant
Nerinetide	FRONTIER	2	LAMS of 2–5 for 15 min or more	3 h	Before hospital admission, before EVT (if performed)	Good functional outcome on a sliding dichotomy of the mRS at day 90	Completed	Overall non-significant, a non-significant trend towards better outcomes observed in patients undergoing reperfusion ≤3 h after symptom onset
NoNO-42: next-generation PSD-95 (non-degradable by alteplase)–NMDA receptor disruption	ACT-42	2	Anterior circulation AIS eligible for IVT with or without EVT; NIHSS >5	3 h	Single intravenous dose before IVT, with or without EVT	Shift in mRS scores at day 90	Active and recruiting	Pending
Cerebral oedema								
Glyburide: SUR1–TRPM4 channel inhibitor	GAMES-RP	2	LVO with DWI-MRI lesion volume of 82–300 cm^3	Within 10 h	Continuous intravenous administration for up to 72 h	mRS score of 0–4 at day 90 without undergoing decompressive craniectomy	Completed	Non-significant reduction in oedema and mortality
Glyburide	CHARM	3	NCCT-ASPECTS of 1–5; or ischaemic core lesion of 80–300 mL on CT perfusion; or DWI-NIHSS ≥10	Within 10 h	Intravenously administered as a bolus followed by 72 h infusion	Shift in mRS scores at day 90	Completed	No significant functional outcome improvement; appeared to show benefit in EVT-treated patients with core <125 mL, however result was non-significant
Oxidative stress, inflammation								
Edaravone dexborneol: free radical scavenger and anti-inflammatory agent	TASTE-SL	3	NIHSS between 6 and 20; prestroke mRS ≤1	48 h	Sublingual administration twice daily for 14 days	mRS score of 0–1 at day 90	Completed	Significantly improved functional outcome
Edaravone dexborneol	TASTE-2	3	NIHSS 6–25; NCCT-ASPECTS 6–10	24 h	Before EVT, twice daily for 14 days	mRS score of 0–2 at day 90	Ongoing	Pending
Edaravone dexborneol	INSIST-ED	3	LVO with mTICI 2b–3; NIHSS ≥6; prestroke mRS ≤1	9 h	Intravenously administered within 60 min after EVT	mRS score of 0–2 at day 90	Completed	No functional outcome improvement
Tocilizumab: anti-IL-6 receptor antibody	IRIS	2	LVO undergoing EVT; NIHSS ≤6	24 h	Single-dose intravenous infusion before EVT	Change in infarct core volume from baseline to 72 h post EVT	Completed	Reduced infarct volume versus placebo, with no safety concerns (preliminary, not peer reviewed)
Edaravone: free radical scavenger	ERASE-STROKE	3	Disabling anterior circulation AIS; non-cardioembolic; not eligible for IVT or EVT	24 h	Oral administration within 24 h after onset, twice daily consecutive 28 days	mRS ≤1 at day 90	Completed	mRS 0–1 at 90 days and NIHSS ≥4 at day 28 significantly improved; mRS 0–2 non-significant
ApTOLL: TLR4 antagonist	APRIL	1b/2a	LVO undergoing EVT; NCCT-ASPECTS 6–10; estimated infarct core on baseline CT perfusion of 5–70 mL	6 h	Before the initiation of EVT	Safety of ApTOLL	Completed	Safe, potential clinical benefit shown at 90 days
Uric acid: free radical scavenger	URICO-ICTUS	2b/3	Undergoing thrombolysis; NIHSS >6 and ≤25; prestroke mRS ≤2	4·5 h	During 90-min alteplase infusion	mRS score of 0–2 at day 90	Completed	No functional outcome improvement

(Table continues on next page)

(NIHSS score 4–25) and had received intravenous alteplase, endovascular thrombectomy, or both. The primary efficacy endpoint was the proportion of patients achieving a favourable functional outcome at 90 days (mRS 0–2). The results showed 56·7% of patients on NBP achieved favourable outcomes versus 44·0% with placebo

	Trial name	Phase	Eligibility criteria	Time window	Administration timing	Primary outcome	Status	Results
(Continued from previous page)								
Multimechanisms								
Butylphthalide	BAST	3	NIHSS 4–25; undergoing rt-PA or EVT or intravenous rt-PA bridging to EVT	6 h	Injection twice daily in the first 14 days and soft 0·2 g capsules 3 times daily for the next 76 days	mRS score of 0–2 at day 90	Completed	Significantly improved functional outcome
Enhancing oxygen and metabolism								
Normobaric hyperoxia: oxygen therapy	OPENS-2	3	LVO undergoing EVT	6 h	Before EVT	Ordinal mRS score at day 90	Completed	Significantly improved 90-day outcomes; no safety concerns
RNS60 (oxygen-enriched saline)	RESCUE	2	LVO undergoing EVT; NIHSS score >5; ASPECTS >5; pre-stroke mRS ≤2	12 h	Immediately post-EVT, 48-hour infusion	mRS score at day 90	Completed, phase 3 ongoing	Safe; fewer deaths, reduced infarct growth; appeared to show better 90-day outcomes, but result was non-significant (proof-of-concept trial)

AIS=acute ischaemic stroke. ASPECTS=Alberta Stroke Program Early CT Score. DWI=diffusion-weighted MRI. EVT=endovascular thrombectomy. IVT=intravenous thrombolysis. LAMS=Los Angeles Motor Scale score. LVO=large vessel occlusion. mCTA=multiphase CT angiography. mRS=modified Rankin Scale. mTICI=modified Thrombolysis in Cerebral Infarction score. NCCT=non-contrast CT. NIHSS=National Institutes of Health Stroke Scale score. PWI=perfusion-weighted MRI. rt-PA=recombinant tissue plasminogen activator.

Table: **Key cytoprotective trials by targeted injury process and agent**

(OR 1·70, significant benefit).[87] Serious adverse events were less frequent in the NBP group and the trial provided robust evidence of both safety and effectivness.[87] However, several factors limit its generalisability: a low percentage of patients received endovascular thrombectomy, many participants had mild strokes (low NIHSS scores) reducing mortality and symptomatic intracerebral haemorrhage rates, and the study was conducted exclusively in China.

Enhancing oxygen and metabolism

Enhancing oxygen delivery in a controlled manner—particularly to hypoperfused yet salvageable penumbral tissue—is a promising strategy to preserve mitochondrial function, limit reactive oxygen species generation during ischaemia, and mitigate reperfusion injury.[88] The RESCUE and OPENS-2 trials explored different oxygen enrichment approaches. RESCUE[89] was a phase 2 trial evaluating the safety of RNS60 (a charge-stabilised, oxygen-enriched nanostructured saline) in patients with large vessel occlusion undergoing endovascular thrombectomy. The study enrolled 82 patients (NIHSS score >5, ASPECTS >5) who were treated within 24 h of symptom onset. RNS60 was infused intravenously starting before arterial access closure and continued for 48 h.[89] The trial showed RNS60 was safe and well tolerated, with 1·0 mL/kg per h reducing infarct growth after endovascular thrombectomy compared with placebo, supporting further efficacy evaluation.[90]

OPENS-2,[91] a phase 2 trial across 26 centres in China, investigated whether normobaric hyperoxia could enhance endovascular thrombectomy outcomes in patients with large vessel occlusion. The trial enrolled 473 patients aged 18–80 years, presenting within 6 h of symptom onset (NIHSS score 10–20, ASPECTS ≥6). Normobaric hyperoxia, delivered via high-flow mask for 4 h around reperfusion, significantly improved 90-day functional outcomes, without increasing adverse events.[91] Subgroup analyses suggested benefits in older patients, those with lower ASPECTS, and those treated 3–6 h after onset.[91] Limitations included ethnic homogeneity, low female representation, and challenges in blinding and oxygen delivery, but the findings support normobaric hyperoxia as a low-cost, scalable endovascular thrombectomy adjunct.

Emerging cytoprotectants

Uric acid, an endogenous antioxidant from purine metabolism, scavenges free radicals, reduces the release of inflammatory factors, protects the vascular endothelium, and stabilises cell membranes and mitochondria.[92] The phase 2b/3 URICO-ICTUS trial[93] enrolled 421 patients treated with alteplase within 4·5 h of symptom onset (NIHSS score >6–25, prestroke mRS ≤2) across ten Spanish centres. Patients were randomly assigned to receive 1000 mg of intravenous uric acid or placebo, infused concurrently with alteplase over 90 min.[93] The trial's primary outcome (mRS score of 0–1) was unmet. However, ordinal analysis of the full mRS distribution showed a shift favouring uric acid (adjusted OR 1·40, 95% CI 0·99–1·98; p=0·055–0·057), and significantly better median mRS scores at 90 days. Exploratory subgroup analyses further suggested possible benefit in women, patients with NIHSS of 10 or less, and those with hyperglycaemia, although these findings were also not statistically significant.[93] More recently, the National Institutes of Health (NIH) Stroke Preclinical Assessment Network confirmed uric acid's cerebroprotective effects across age, sex, and comorbidities in a diverse multicentre rodent model.[94] Uric acid is now being evaluated in the NIH Stroke Preclinical to Clinical Evaluation Platform for further clinical translation.

APRIL[95] (phase 1b/2a) evaluated ApTOLL, a selective Toll-like receptor 4 antagonist that aims to suppress sterile inflammation triggered by ischaemia–reperfusion.

In 151 endovascular thrombectomy-treated patients enrolled within 6 h of symptom onset, ApTOLL was well tolerated, with no increase in symptomatic intracerebral haemorrhage or mortality, and showed significant reduction in infarct growth and better functional outcomes in the high-dose group.[96] Post-hoc analyses further confirmed that ApTOLL 0·2 mg/kg reduced final infarct volume, infarct growth, and cerebral oedema at 72 h after symptom onset, with the greatest benefit observed in those with large cores, fast progressors, and incomplete reperfusion.[97] Larger trials are needed to validate these findings.

Future trial considerations

The timing of intervention is pivotal for the success of cytoprotective strategies for acute ischaemic stroke. Therapies should be administered within a therapeutic window that aligns with their physiological target—a principle clearly shown in nerinetide trials.[71,72] Nerinetide showed encouraging results only when administered prehospital within 3 h of symptom onset,[72] suggesting that therapies targeting excitotoxicity are most effective early in the ischaemic cascade (figure 4). In contrast, administering such therapy too close to rapid and complete reperfusion, as in ESCAPE-NEXT[69]—provides too narrow a window to test the drug's effect, as excitotoxic processes are already resolving. When determining the appropriate treatment window, it is important to consider individual patient physiology rather than applying a rigid temporal window.[98] Biological heterogeneity, particularly between fast and slow progressors, means tissue injury can vary significantly between patients (figure 3B, C).[99,100] Patients with good collaterals and slower infarct growth might already have much of their tissue preserved at reperfusion, leaving little room to detect added benefit.[69] Enriching early phase or middle-phase studies with patients exhibiting rapid infarct growth could, therefore, improve proof-of-concept sensitivity.

Imaging-based approaches such as perfusion–diffusion mismatch, core–penumbra analysis, and collateral assessment can help identify salvageable tissue. Nonetheless, concerns remain regarding the use of different imaging modalities in large trials, particularly in terms of generalisability and accessibility across diverse health-care settings.[11] Ongoing debates include the definitions and measurements of the infarct core and penumbra, thresholds for defining a large infarct core, and distinctions between perfusion and clinical–core mismatch.[101] Trials using both CT and MRI should anticipate increased variability, as seen in the CHARM trial, where heterogeneous imaging criteria introduced variation in infarct core estimation.[77]

Clinical trial endpoints must be sensitive to their expected biological effects to accurately assess drug efficacy.[102] Although the 90-day mRS score remains essential, it might not fully capture the impacts of cytoprotection. Different mechanisms require tailored outcome measures: anti-oedema agents might be best assessed with metrics such as midline shift or oedema volume; agents targeting excitotoxicity might benefit from endpoints such as infarct volume reduction; and anti-inflammatory therapies could be evaluated with cytokine biomarkers or inflammation-specific scales. Supplementing mRS scores with mechanism-relevant secondary outcomes can reveal meaningful biological effects in early phase studies. Moving beyond static endpoints to target dynamic, early phase processes, such as infarct reduction over time could further improve assessment. Nonetheless, showing improvement in functional outcomes will be essential for adopting drugs in clinical practice.

Practical logistics also matter for the success of adjunctive therapy trials in acute stroke. Fast reperfusion remains the priority in acute stroke care. System-level strategies, such as direct transport of patients with large vessel occlusion to comprehensive stroke centres, deployment of mobile stroke units, and prehospital intravenous thrombolysis via telemedicine, can further shorten onset-to-treatment times.[103,104] For cytoprotective therapies that can be initiated before arrival at a comprehensive stroke centre, establishing hub-and-spoke enrolment systems, where drugs are initiated at referring community hospitals before transfer, can facilitate earlier intervention and expand trial access to diverse populations.

Ultimately, clinical translation begins long before the first patient is enrolled in a trial. The design and rigour of preclinical studies form the foundation for all subsequent progress. Initiatives such as Stroke Therapy Academic Industry Roundtable[105] and the Stroke Preclinical Assessment Network[106] emphasise the importance of multicentre collaboration, rigorous blinding, adequate sample sizes, and the use of relevant animal models with multimodal outcome assessments. Furthermore, phase 2 trials should show biological activity through imaging-based or biomarker-based endpoints, incorporating clinical endpoints to detect early efficacy signals and establish optimal dosing before progressing to large, outcome-driven phase 3 trials.

Conclusions and future directions

The successful outcome of future adjunctive therapies hinges on a multifaceted approach that integrates timely administration, biologically informed patient selection, and tailored endpoints. Early signs from multitarget agents[81,88] suggest that interventions acting on several pathways hold promise, but proof of benefit is yet to be shown.

Recent studies have emphasised the importance of genetics in drug discovery,[107] showing that drug–target pairs supported by genetic evidence have a substantially higher likelihood of gaining approval compared with those without genetic evidence.[108] As more targeted and multimodal adjunctive therapies for acute ischaemic stroke are developed, multiomic and

Search strategy and selection criteria

References for this Review were identified by systematic searches of PubMed, Google Scholar, and ClinicalTrials.gov, focusing on studies published between June 19, 2020, and June 19, 2025. Additional relevant studies published before Sept 18, 2025, were included following reviewer suggestions, to ensure the inclusion of the most recent evidence. The search terms included: "stroke", "cerebrovascular accident", "acute stroke", "acute ischemic stroke", "large vessel occlusion", "endovascular therapy", "mechanical thrombectomy", "endovascular thrombectomy", "endovascular treatment", "EVT", "neuroprotection", "cytoprotection", "reperfusion injury", "intra-arterial thrombolysis", "ischemia–reperfusion injury", "cerebral oedema", "brain oedema", "hemorrhagic transformation", "secondary injury", "delayed cerebral injury", "inflammatory response", "oxidative stress", "treatment outcome", "excitotoxicity", and "acute stroke clinical trials". ClinicalTrials.gov was searched with the terms "ischemic stroke" (condition or disease) and "endovascular therapy" (intervention or treatment), with a particular focus on trials enrolling patients with large vessel occlusion. Additional references were identified through expert recommendations, relevant bibliographies, and the personal databases of the authors. We included peer-reviewed clinical trials, observational studies, meta-analyses, systematic reviews, and translational preclinical research with direct relevance to large vessel occlusion, endovascular thrombectomy, ischaemia–reperfusion injury, and potential adjunctive strategies. References were selected based on relevance, quality, and alignment with the Review's scope. There were no language restrictions. We excluded publications that lacked relevance to acute ischaemic stroke, ischaemia–reperfusion injury, neuroprotection or cytoprotection, or adjunctive reperfusion strategies; were not original peer-reviewed research, systematic reviews, meta-analyses, or substantive translational studies (eg, letters or conference abstracts without full data); focused solely on haemorrhagic stroke, transient ischaemic attack, or non-cerebrovascular conditions; or were duplicate reports of the same dataset.

plasma-pharmacodynamic biomarkers that can bridge mechanistic insights with clinical outcomes will be needed. These biomarkers are essential not only for finding the targets, but also for understanding how and when a therapy exerts its effect across diverse patient populations. Multiomic platforms, such as single-cell transcriptomics, proteomics, and metabolomics, make it possible to holistically understand the cellular and molecular biology underlying ischaemic-reperfusion injury.[109]

Looking ahead, many promising therapies remain under investigation. Whether these encouraging early phase results will translate into clinical benefit remains to be seen. Successful therapies will depend on aligning the biological target with the appropriate time window and selecting mechanism-specific intermediate endpoints for the relevant patient subgroup. Universal designs are unlikely to succeed in such a complex and heterogeneous disease.

Contributors

MG, JL, and KNS researched data for the Review. MG and KNS conceptualised the manuscript. MG wrote the first draft. All authors contributed substantially to the discussion of the content and reviewed and edited the manuscript before submission.

Declaration of interests

NS receives research support from NIH and is a cofounder and shareholder in Bexorg. GSS receives research support from the Brazilian Ministry of Health (for the Optimal Stroke Study); received consulting fees, honoraria for lectures and presentations, and support for travel from Boehringer Ingelheim and AstraZeneca; and is on an advisory board for AstraZeneca. WTK receives research support from NIH, Alzheimer's Association, Hyperfine; has received consulting fees from Astrocyte Pharmaceuticals and Grace Therapeutics; and holds stock options in Woolsey Pharmaceuticals and Grace Therapeutics. KNS received research support granted to Yale University from NIH, American Heart Association, Hyperfine, and Genentech; fees from Sense and Phillips for data safety and monitoring board committee service; consulting fees from Astrocyte, Bexorg, BrainQ,and Rhaeos; and is a cofounder of Alva Health. TNN reports receiving speaker fees from Genentech, Kaneka, and Route92; serving on the SELECT2 and TESLA data safety and monitoring board (unpaid); receiving payment for participation on the CREST-2 data safety and monitoring board; and serving as president of the Society of Vascular and Interventional Neurology (unpaid) and associate editor for *Stroke* (paid).

Acknowledgments

We thank Ivo Bach for providing the angiographic images included in this review as an illustrative example of successful EVT in large vessel occlusion.

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